BEFORE YOU READ

Two forms of a probability function

A random variable is described by a PMF when it is discrete and by a PDF when it is continuous. The two look alike but mean different things.

Probability function

2 forms

PMF $p(x) = P(X = x)$ discrete	**PDF** $f(x)$ continuous
Meaning: each value has its own probability	Meaning: a density, not a probability
Total: values sum to 1	Total: area under the curve is 1
Examples: a die, number of heads	Examples: waiting times, heights

TAKEAWAY

Heights mean different things

A PMF height is a probability; a PDF height is a density, and probability is the area under it.